

Descriptor: *Underwater Stereo Vision Dataset for Transparent Object Detection & Ranging (UW-TransStereo)*

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ABSTRACT Robust detection and ranging of transparent debris in water is challenging due to refraction, low contrast, backscatter, and turbidity. We present a paired, multi-environment stereo dataset acquired with a stereo camera, comprising 25 recordings totalling over 9,000 stereo frame pairs of three bottle targets (two transparent plastic bottles with filled/unfilled variants and a brown glass medicine bottle) across air and four underwater conditions (freshwater, freshwater with enhancement, saltwater, and saltwater with suspended pellets), along with analysis-ready detector spreadsheets and scripts. To promote reproducible use, we provide a standard evaluation protocol with well-defined accuracy measures—mean absolute error (MAE), root mean square error (RMSE), coefficient of determination (R^2), and mean signed error (bias)—and report the proportion of frames with valid estimates (“valid %”), including vertical region-of-interest (ROI) sampling (upper/centre/lower). We include a lightweight vision enhancement variant, “basicEnhance”, to illustrate how pre-processing can be benchmarked against the same protocol without retraining models. Users can load the paired dataset files to benchmark alternative detectors or compare ranging strategies across the five environmental conditions. The release is intended as a neutral testbed for benchmarking underwater transparent-object perception and for stress-testing stereo ranging under refractive media. File naming, storage structure, and missing-value conventions (including explicit use of not-a-number cases) are documented to ease integration. We release all raw stereo streams, detector outputs, qualitative screenshots, and analysis scripts.

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DATA DOI/PID 10.5281/zenodo.16753748

DATA TYPE/LOCATION Stereo video and detector spreadsheets; London, U.K. (lab tank)

INDEX TERMS computer vision, object detection, refractive media, stereo vision, transparent objects, underwater imaging

BACKGROUND

Underwater perception of transparent objects remains difficult: refractive index matching between object and medium suppresses appearance cues; turbidity and backscatter degrade contrast; and specularities violate brightness constancy [1]. This *index-matching* effect is particularly severe for water-filled transparent containers: because glass (refractive index $n \approx 1.52$) and water ($n \approx 1.33$) have similar refractive indices [2], [3], [4], the optical contrast at the container wall is greatly reduced, causing stereo matchers to

see through the target and lock onto background surfaces—producing rear-wall depth estimates or invalid (NaN) returns. While stereo geometry can help, practical performance varies across environments and targets.

Prior work on transparent-object perception and underwater imaging shows that while stereo brings geometric constraints, its performance deteriorates with scattering and contrast loss; most public datasets emphasise opaque targets or a single capture regime [5]. Our release complements these efforts by focusing on transparent bottles across multi-

ple media and visibility regimes and by furnishing stratified metrics and environment-aware guidance to support robust benchmarking. Our earlier work analysed freshwater *day* sequences [5]. In this Descriptor, we release the paired *night* recordings and use the prior day results for contextual comparison in the time-of-day analysis.

This article describes and releases a multi-environment stereo dataset targeted at transparent bottles and related targets, with matched detector outputs and analysis-ready summaries. The data are designed to support benchmarking under domain shift (air↔water, clear↔turbid) and to quantify how often stereo delivers valid ranges versus failure modes (e.g., NaN or rear-surface estimates). Users can load the paired .svo2/.xlsx files to benchmark alternative detectors or compare ranging strategies across the five environmental conditions. We also summarise our previously reported feasibility and analysis studies to position the value of this release and to aid reuse [5], [6], [7]. Unlike related public datasets that emphasise opaque targets or single-condition tests, our release focuses on transparent bottles under multiple media to support robustness studies and cross-domain generalisation.

COLLECTION METHODS AND DESIGN

Sensor and rig. A calibrated ZED Mini stereo camera is mounted to a windowed tank rig (approx. H30×W26×D116 cm) (Fig. 1) with a blue cloth attached to the rear wall. Trials are conducted in air and water (fresh and salt), including suspended plastic pellets. Distances span a practical range for small remotely operated vehicle (ROV) tasks; air serves as the best-case baseline.

Targets. We use a smooth and a textured transparent plastic bottle, including filled with water and unfilled variants, and an opaque (control) small brown glass medicine-bottle (Fig. 2), to probe how surface texture and fill state affect detectability and stereo ranging across the environments (Fig. 3).

Calibration and computational steps (explicit). We perform standard stereo calibration and rectification using the ZED SDK [8] prior to all recordings; depth is computed in NEURAL mode with fixed parameters across sessions. Each session executes the same steps that yield the released files:

- 1) acquire rectified stereo frames (.svo2);
- 2) *optional*: apply enhancement, e.g. *basicEnhance* (edge emphasis + confidence reweighting);
- 3) run YOLOv8 inference on the left stream;
- 4) sample stereo range at three vertical ROIs (Upper/Centre/Lower) per accepted detection;
- 5) log per-frame measurements to spreadsheets and save qualitative screenshots.

These steps correspond to the blocks in Fig. 4 and to the file types enumerated in *RECORDS AND STORAGE*.

Acquisition protocol. For each environment–target combination, we record paired stereo streams (.svo2 format)

and run a fixed detection pipeline to export per-frame detections to spreadsheets. For each accepted detection, stereo range is sampled at three vertical ROI bands (Upper/Centre/Lower) within the bounding box. Each sample is associated with environment, bottle type and detector confidence, and compared to independently measured ground-truth distance marks.

Detections pipeline. Unless otherwise stated, we run inference with a YOLOv8 model configured at 416×416 px image size, a confidence threshold of 0.40, and an IoU threshold of 0.45; depth estimation employs the ZED NEURAL mode throughout. All detections are performed on the left stereo stream only, with no multi-view or temporal fusion in the baseline pipeline. *Overview.* The end-to-end workflow—from stereo calibration and depth to detection, logging, and artefact generation—is summarised in Fig. 4.

Enhancement variant To enable method comparisons without retraining, we include a minimal pre-processing variant, *basicEnhance* (edge emphasis via Canny+dilation blended with RGB; depth-aware confidence reweighting at the box centre). Parameters are fixed for reproducibility.

VALIDATION AND QUALITY

To ensure validity and reliability, we collected multiple observations at each marked distance across repeated trials per environment, yielding a sufficient sample for per-environment aggregation; the exact counts and confidence intervals are regenerated by the supplied scripts.

Metrics and aggregations. We compute mean absolute error (MAE) and root mean squared error (RMSE) (cm) for distance estimates, the coefficient of determination (R^2) against ground truth, and the mean signed error (bias). *Definition.* The valid-range rate (“valid %”) denotes the percentage of detection samples for which the stereo system returns a finite depth value—i.e., frames where ranging succeeds rather than returning NaN (not-a-number). We report valid % (percentage of frames with a valid estimate), stratified by environment, bottle type and vertical ROI. Environment-level baseline values (centre ROI; detector settings per text) are summarised in Table 1.

Observed baseline patterns. Air establishes the expected best-case performance. Underwater conditions increase error and reduce valid % relative to air; suspended pellets and low visibility exacerbate both. Water-filled transparent bottles are the hardest cases, frequently producing NaN or rear-surface estimates when the object is near index-matched to the medium. *Per-bottle overview.* A comprehensive set of performance heatmaps (distance root mean squared error (RMSE) and classification/confidence metrics), stratified by environment and bottle type, is provided in the Supplementary Material (Fig. S1) to orient users to condition- and target-specific behaviour.

Confidence tends to decrease with object distance and co-varies with range error. However, calibration weakens in high backscatter, cautioning against naive confidence gating.

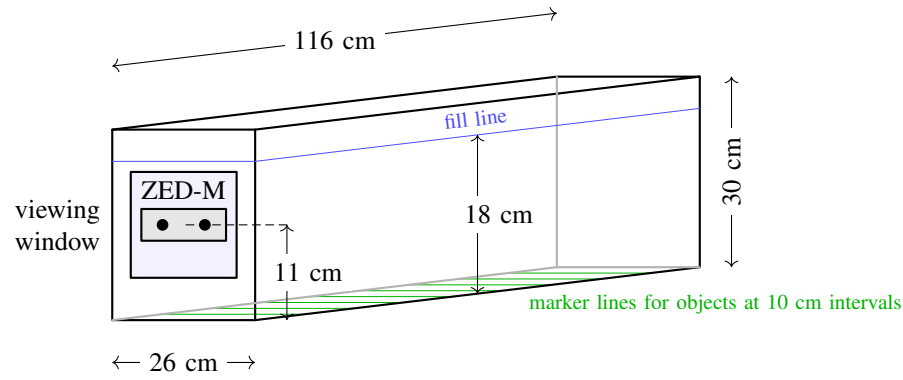


FIGURE 1. Schematic of the experimental tank showing perspex viewing window and ZED Mini mounting, the water fill line, and marker lines added to the base of the tank for object placements at 10 cm equidistance.

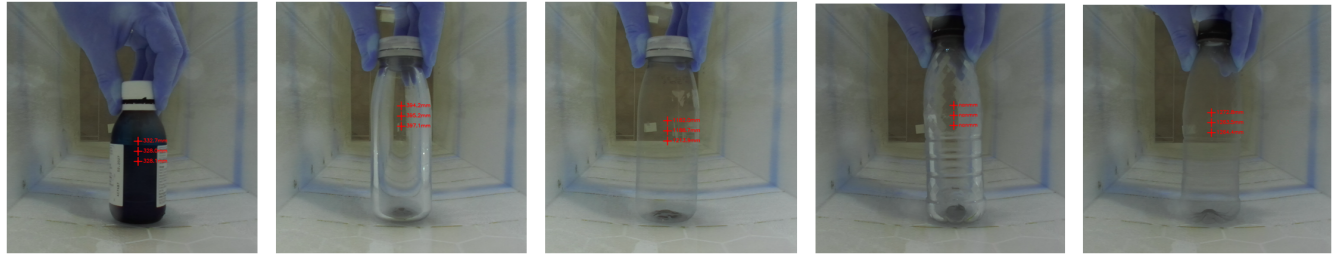


FIGURE 2. Bottles used in the experiments, from left to right: medicine bottle, smooth bottle, smooth bottle filled, textured bottle, textured bottle filled.

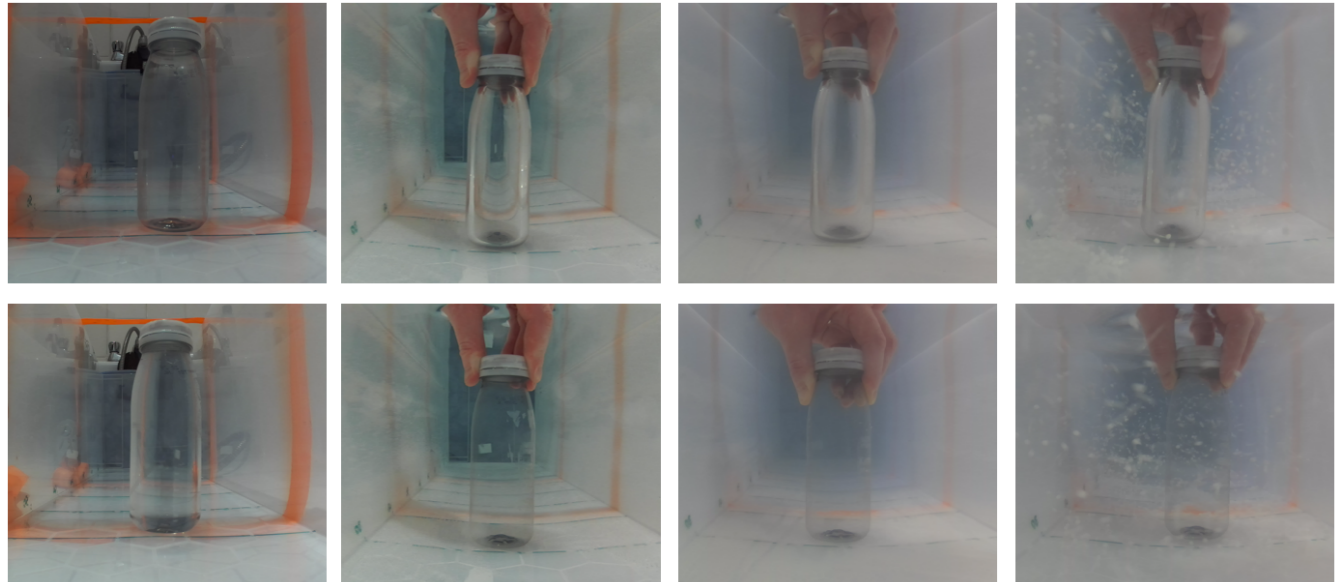


FIGURE 3. Smooth clear bottle air-filled (top) and water-filled (bottom), shown across the multi-environment set (left to right): air, freshwater, saltwater, saltwater with pellets.

Across the underwater media, errors are highest in Freshwater and remain elevated in Saltwater and Saltwater-pellets relative to Air, with suspended pellets further degrading valid % especially for filled transparent bottles. These trends across the released environments can be regenerated from the provided spreadsheets.

Illustrative pre-processing (basicEnhance). To enable method comparisons without retraining, we include a min-

imal inference-time variant that (i) emphasises edges via Canny followed by $2\times$ dilation and blends them with the RGB input, and (ii) applies depth-aware confidence reweighting at the box centre:

$$\text{conf}' = \text{conf} \cdot \frac{1}{1 + d_{\text{mm}}/1000},$$

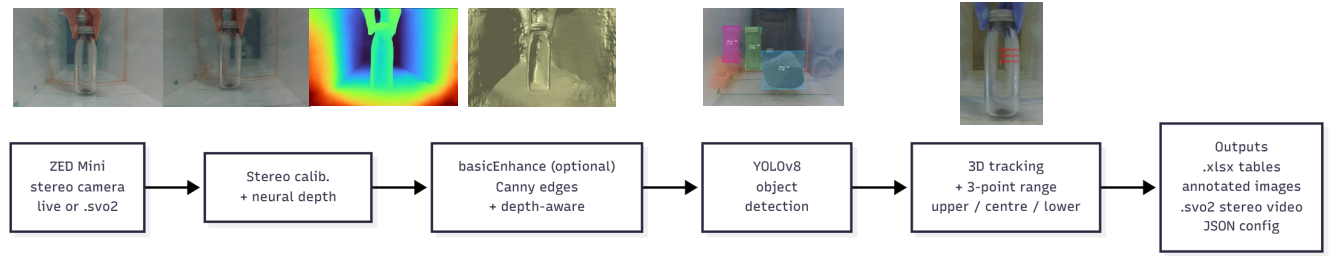


FIGURE 4. End-to-end processing pipeline: stereo calibration/depth → optional edge/depth-aware confidence → YOLOv8 detection → ZED ingestion/tracking → per-detection measurements and logs (Excel, screenshots, SVO, config). Accompanying screenshots are included as examples of the data alongside the processing steps.

TABLE 1. Baseline technical quality by environment: original unbalanced data vs. distance-stratified normalised data (RMSE_{norm}: 10 samples per bottle type per distance, 200-800mm range, includes failed detections as zeros). MAE and RMSE calculated from median deviation of valid measurements; valid % shows detection success rate; #valid shows absolute count of successful measurements.

Environment	MAE (mm)	RMSE (mm)	valid %	#valid	RMSE _{norm} (mm)
Air	179.9	204.7	95.7	902	225.5
Freshwater	273.0	327.8	70.5	473	530.3
Freshwater + enhance	246.4	299.5	88.3	484	456.2
Saltwater	175.8	247.6	70.3	526	365.8
Saltwater + pellets	183.5	250.8	70.8	235	385.6

where d_{mm} is the centre-depth in millimetres. All parameters are fixed across runs for reproducibility. Quantitative ablation results for the freshwater subset were previously reported in [5]; here we release the data and scripts needed to reproduce and extend those analyses across all environments. Users can examine the trade-off between improved valid-range rate (“valid %”) and potential increases in distance error under the same protocol and thresholds; tables/figures and regeneration scripts are provided (see *SOURCE CODE AND SCRIPTS* and Supplementary).

RECORDS AND STORAGE

Scope and composition. The archive contains 25 paired recordings (50 core files) across *Air*, *Freshwater*, *Freshwater-basicEnhance*, *Saltwater*, and *Saltwater-pellets*, plus 216 qualitative screenshots (see Table 2). Collated analysis tables contain >5,000 labelled measurements across environment–bottle combinations (frames, detections, ROI samples). *Note on time-of-day:* For freshwater, the released raw videos correspond to *night* sequences; the *day* sequences used for the day–night comparison were part of our earlier study, which did not record the .svo2 data; hence, are not included in this archive.

Modalities and file types. Each paired recording includes the following artefacts:

- **Raw stereo video:** .svo2 (ZED HD2K, 15 FPS, H.264).
- **Detector outputs:** .xlsx per recording with per-frame detections (class, confidence, bounding box).
- **Reference visuals:** .png screenshots for qualitative checks.

Organisation and naming. Files are flat-structured with context in the name: [Environment]_[BottleType]_[filekind]_[timestamp].[ext] e.g., Saltwater-pellets_smtBtl-fill_detections_08052025_183311.xlsx. Matching timestamps pair each .svo2 with its .xlsx; screenshots mirror their parent. A typical run yields paired .svo2/.xlsx plus screenshots and a configuration snapshot, consistent with the pipeline in Fig. 4. An excerpt of the archive manifest (paths, types, purposes) is provided in the Supplementary Material (Table S1); the full machine-readable manifest is supplied alongside the dataset release.

Data records (schema). Each measurement row includes: ground-truth distance; stereo-derived distances sampled at upper/centre/lower ROIs; detector confidence; environment and bottle identifiers; and time/context fields where available. Missing values are retained (e.g., NaN when the target cannot be ranged). Units are documented per column in the data dictionary. Selected columns are listed in the Supplementary Material (Table S2); the full data dictionary is supplied alongside the dataset release.

Availability and licensing. All files (paired .svo2/.xlsx, screenshots, collated CSVs, per-condition summaries, and scripts) will be deposited with a DOI (CC BY 4.0 licence; final licence string confirmed in the deposit record).

Archive manifest. An excerpted example manifest and the full machine-readable manifest mapping each recording_id to paired .svo2/.xlsx files are provided in the Supplementary Material (Table S1).

Column dictionary. A selected column dictionary for the collated CSV files is provided in the Supplementary Material (Table S2), with the full column dictionary supplied alongside the dataset release.

TABLE 2. Dataset summary by environment.

Environment	#Recordings	#Frames	#Detections	#Screenshots
Air	5	1,654	978	35
Freshwater	5	1,980	914	45
Freshwater- -basicEnhance	5	1,720	1,044	45
Saltwater	5	1,864	768	45
Saltwater-pellets	5	1,873	460	46
Total	25	9,091	4,164	216

INSIGHTS AND NOTES

Intended use. The dataset supports cross-domain generalisation (air $\leftrightarrow$ water; clear $\leftrightarrow$ turbid), enhancement ablations (e.g., *basicEnhance*), as first analysed in [5], error-mode analysis (e.g., rear-surface estimates), and benchmarking of detection+ranging pipelines from raw streams or spreadsheets.

Practical guidance. Environment- and target-aware strategies—e.g., selecting the more stable vertical ROI, applying mild pre-processing, or gating by confidence—can improve usable estimates without retraining. The provided protocol enables systematic comparisons of such choices.

Limitations. Results inherit the behaviour of the chosen detector and ZED mode; conditions are diverse but not exhaustive (e.g., extreme turbidity, dynamic lighting). Transparent objects beyond bottles may differ.

Ethical considerations. Data were collected in controlled settings using inanimate targets; no personal data or wildlife were involved. Please cite both this article and the dataset DOI when reusing.

Secondary uses. Beyond detection + ranging benchmarks, the paired confidence and error traces can support (i) confidence calibration studies under refractive domain shift, and (ii) robustness profiling across controlled turbidity gradients.

SOURCE CODE AND SCRIPTS

Reproducibility. We release a code repository with pinned dependencies and one-command scripts to regenerate all tables and figures in this article from the released comma-separated values (CSV) files. The repository includes: (i) example code for reading .svo2 stereo recordings and parsing detector spreadsheets, (ii) scripts that reproduce the environment–bottle aggregations and heatmaps reported here, and (iii) a machine-readable manifest mapping each recording_id to its paired .svo2/.xlsx files. Supplementary material (PDF, co-hosted with the code release) provides extended plots and analyses and a full column dictionary for downstream reuse.

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A.L.S. developed the acquisition protocol and experimental rig, curated and analysed the data, and wrote the manuscript. C.O. and I.F. supervised the work and provided feedback on the dataset design, analysis, and manuscript. All authors reviewed the manuscript.

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